

General insurance reserving and claims triangle analysis

Development methods and reserve uncertainty

A comparison of paid and incurred reserve estimates, linking claims development patterns to method selection, retrospective accuracy and uncertainty.

Methods	R; Chain Ladder; Bornhuetter–Ferguson; Mack uncertainty estimates
Data	CAS Schedule P, private passenger auto; accident years 1998–2007
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1 Executive summary

This project uses CAS Schedule P private passenger auto data to construct paid and incurred claims development triangles and estimate unpaid claims reserves. The portfolio contains 13,250 Schedule P records across 143 companies or groups, with accident years from 1998 to 2007 and development lags from 1 to 10. The data were aggregated to a portfolio-level view and censored at the 2007 valuation date, giving a 55-cell observed reserving triangle.

The paid Chain Ladder estimate produced a total ultimate loss of 157.7m and a reserve estimate of 18.9m. This was very close to the retained 10-year paid outcome of 157.7m. The Bornhuetter-Ferguson paid estimate was more conservative, giving an ultimate estimate of 160.1m and a reserve of 21.3m. This reflects the expected loss ratio prior placing more weight on remaining unpaid development for immature accident years, especially 2006 and 2007.

The incurred basis gave a different view. Both Chain Ladder and Bornhuetter-Ferguson projected a small reduction of around 0.7m from the latest incurred position. This is development relative to existing incurred estimates, not a negative amount of unpaid claims. This is consistent with the incurred development pattern, where several accident years showed modest downward revisions as case and bulk reserves were updated.

Mack Chain Ladder gave a paid reserve of 18.9m with an approximate 95% interval from 18.2m to 19.7m. On the incurred basis, the Mack interval ranged from -1.5m to 0.1m, crossing zero and supporting the interpretation that the incurred reserve position was close to fully developed at the valuation date.

Overall, the paid basis gives the clearest view of remaining settlement development, while the incurred basis provides a useful comparison against a more mature reserve view. The results are coherent for a private passenger auto portfolio: paid losses develop strongly in the early lags, incurred losses are more stable, and reserve uncertainty is concentrated in the most recent accident years.

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2 Data and scope

The project uses loss reserving data published by the Casualty Actuarial Society. The data are based on NAIC Schedule P claims records and were prepared for claims reserving studies across major U.S. property-casualty lines of business [1]. Schedule P data are suitable for this type of project because they organise paid and incurred claim development by accident year and development period.

This report focuses on the private passenger auto line of business. The line was selected because its paid development is relatively clear and intuitive compared with longer-tailed classes such as workers' compensation, medical malpractice and products liability. The main fields used are accident year, development year, development lag, cumulative paid losses, cumulative incurred losses, bulk loss and earned premium.

The source records were aggregated to a single portfolio-level triangle by accident year and development lag. The valuation year was set to 2007, and a cell was retained in the observed triangle only if

$$\text{Accident Year} + \text{Development Lag} - 1 \leq 2007.$$

This creates the standard upper-left observed triangle: the 1998 accident year has ten observed lags, the 1999 accident year has nine, and the 2007 accident year has one. All monetary amounts are divided by one million from the source data scale and labelled m. This scaling should not be confused with a separately verified currency denomination.

Table 1. Portfolio data summary.

Metric	Value
Line of business	Private Passenger Auto
Valuation year	2007
Schedule P records	13,250
Companies or groups	143
Accident years	1998–2007
Development lags	1–10
Observed triangle cells	55
Expected observed cells	55
Latest observed paid	138.8m
Latest observed incurred	159.7m
Net earned premium	220.2m
Actual 10-year paid	157.7m
Actual 10-year incurred	158.7m

Figure 1 shows the observed cumulative paid triangle at the 2007 valuation date. The triangular structure confirms that future development has been excluded from the modelling view.

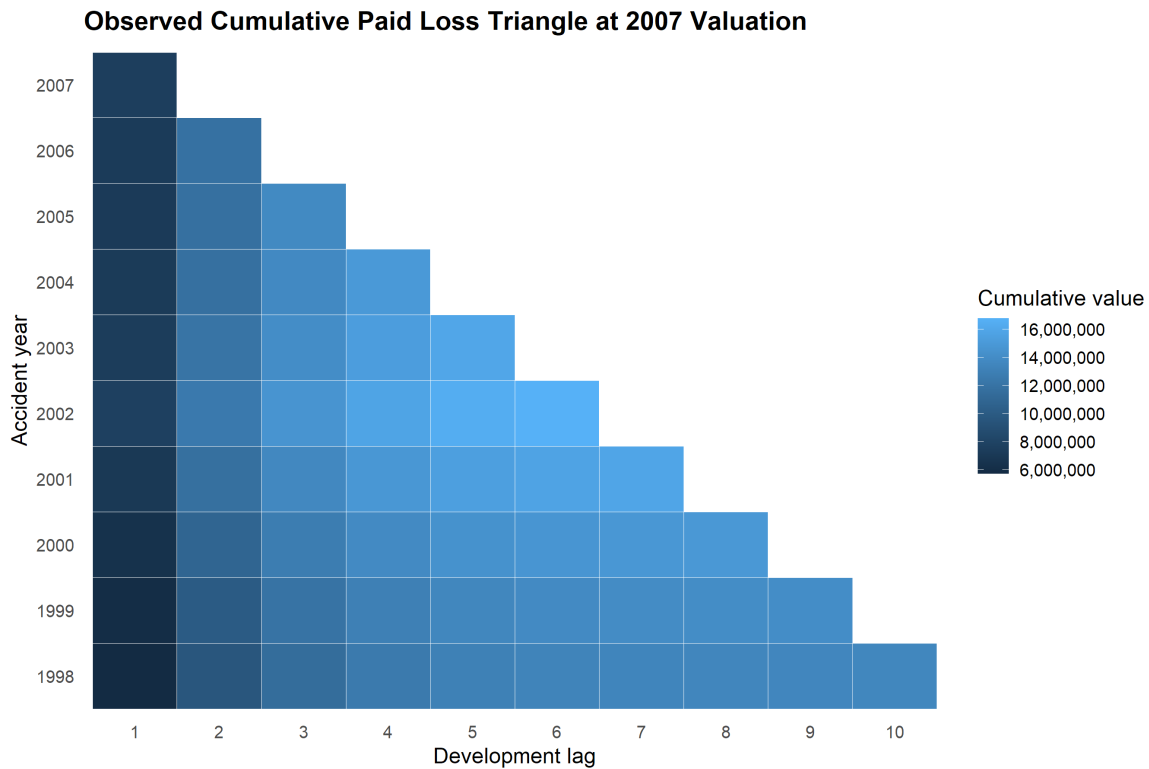


Figure 1. Observed cumulative paid loss triangle at the 2007 valuation date.

3 Methodology

3.1 Chain Ladder

Chain Ladder is fitted separately to the paid and incurred triangles, using volume-weighted age-to-age link ratios. The cumulative development factor projects the latest observed value to ultimate. The reported reserve is the difference between estimated ultimate and the latest observed value.

On the paid basis, that difference estimates remaining unpaid development. On the incurred basis, it estimates changes to the existing incurred position, which already includes case and other reserve estimates. A negative incurred result therefore represents an expected release; it does not imply that unpaid claims are negative.

3.2 Bornhuetter–Ferguson

The Bornhuetter-Ferguson method was used as a comparison against the pure Chain Ladder estimates [5]. The method combines the latest observed loss position with an expected ultimate loss prior based on earned premium and a selected expected loss ratio. The expected loss ratio was selected from older observed incurred-to-premium ratios, giving a prior of approximately 77.5%.

For an accident year with latest observed loss L , expected ultimate loss E , and Chain Ladder factor to ultimate F , the Bornhuetter-Ferguson estimate is

$$\text{BF Ultimate} = L + E \left(1 - \frac{1}{F} \right).$$

For immature accident years, where only the first one or two development lags are observed, the Chain Ladder ultimate is highly sensitive to early paid development. Blending the observed loss position with a stable expected loss ratio prior makes the Bornhuetter-Ferguson estimate less dependent on sparse early development data.

3.3 Mack uncertainty estimates

Mack Chain Ladder was applied to quantify uncertainty around the Chain Ladder reserve estimate. The method is a stochastic extension of Chain Ladder and estimates standard errors without requiring a full parametric distribution for claims development [4]. The implementation uses the R `ChainLadder` package [3].

The approximate intervals are reported around the estimated reserve. They depend on the Chain Ladder assumptions and are not a complete distribution of all reserve risks.

4 Results and discussion

4.1 Chain Ladder estimates

The selected paid development factors are highest at early lags and converge towards one as claims settle. Incurred factors are closer to one and fall slightly below one at some lags, indicating downward revisions to existing incurred estimates.

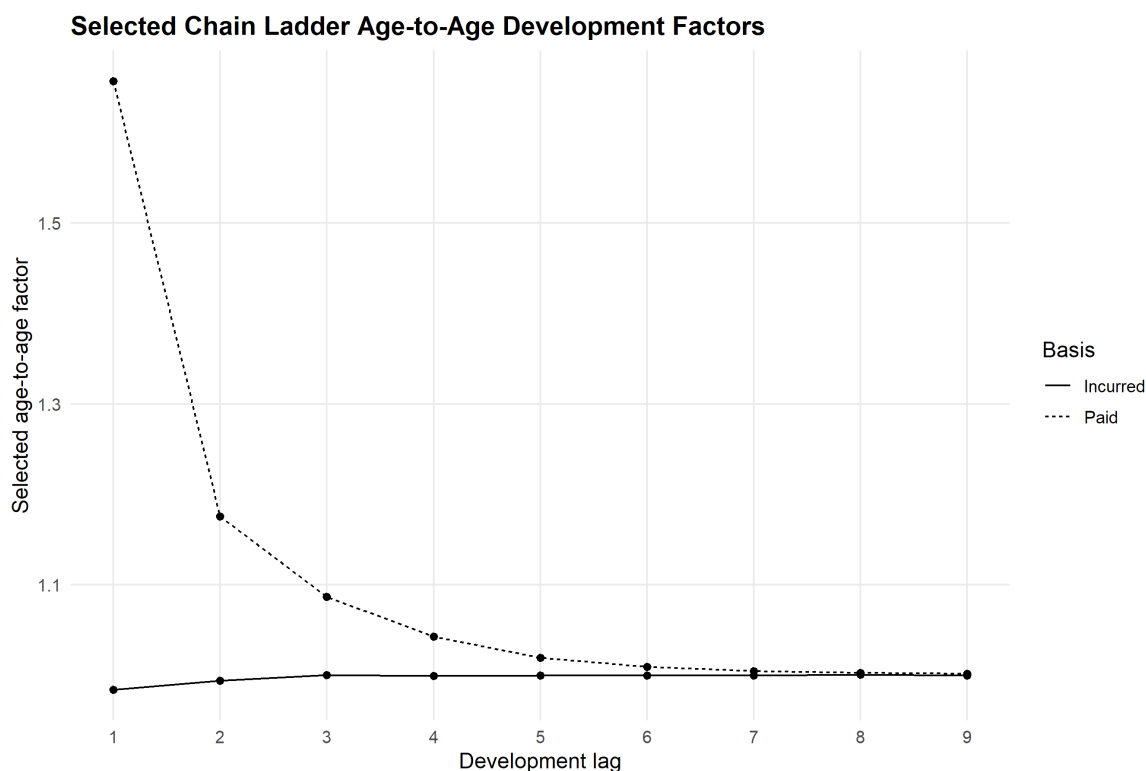


Figure 2. Selected Chain Ladder age-to-age development factors.

The paid Chain Ladder estimate was very close to the retained 10-year paid outcome, with a total error of approximately -0.02m. This should not be interpreted as proof that the method is generally

Table 2. Chain Ladder summary.

Basis	Latest observed	Ultimate	Reserve	Actual 10-year	Error
Paid	138.8m	157.7m	18.9m	157.7m	-0.02m
Incurred	159.7m	159.1m	-0.7m	158.7m	0.34m

exact, but in this retrospective backtest it performed well for the aggregate private passenger auto triangle. The aggregation across 143 companies contributes to this stability: idiosyncratic development patterns at company level tend to diversify, producing smoother volume-weighted link ratios than would be observed in a single insurer's triangle.

The incurred Chain Ladder estimate produced a small negative reserve. This is reasonable in context: incurred losses already include reserve estimates, and the observed incurred development shows some downward revision over time. The incurred result therefore reflects a small implied release rather than additional outstanding development.

4.2 Bornhuetter–Ferguson comparison

The resulting reserve estimates are shown in Figure 3. Paid reserves are concentrated in the most recent accident years. Bornhuetter-Ferguson is more conservative than Chain Ladder on the paid basis, particularly for the 2006 and 2007 accident years, because the expected loss ratio prior gives more weight to remaining unpaid development where paid development is immature.

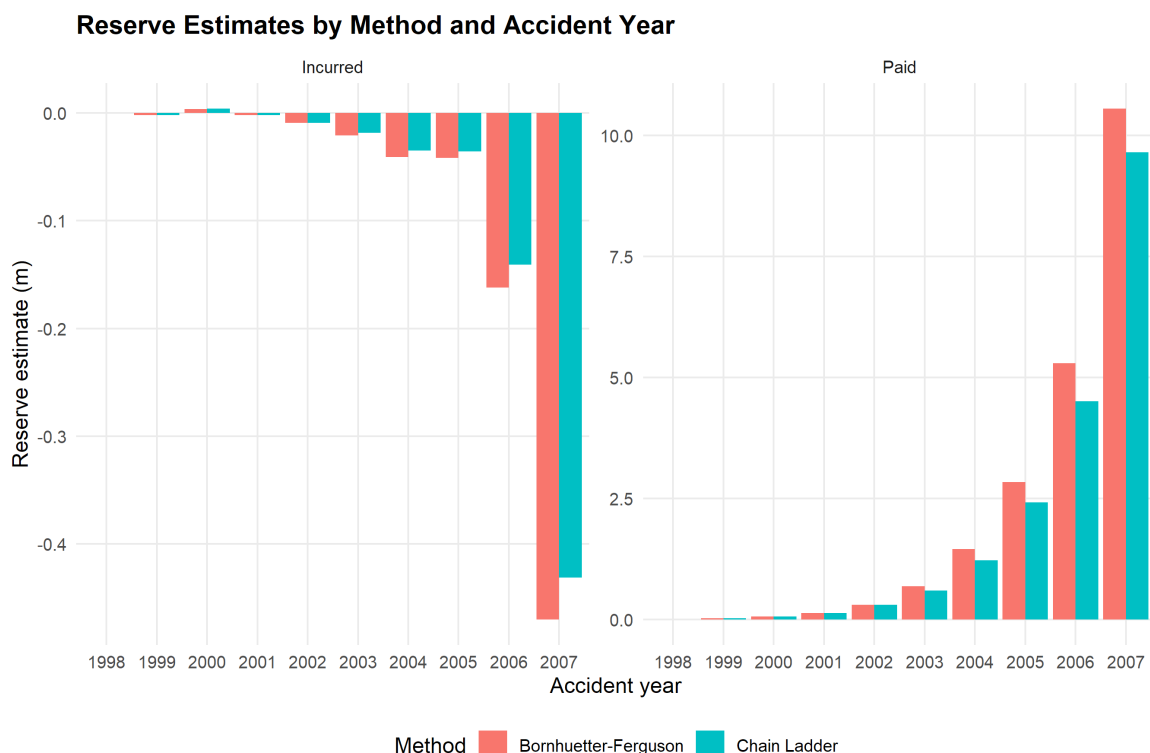
**Figure 3.** Reserve estimates by method and accident year.

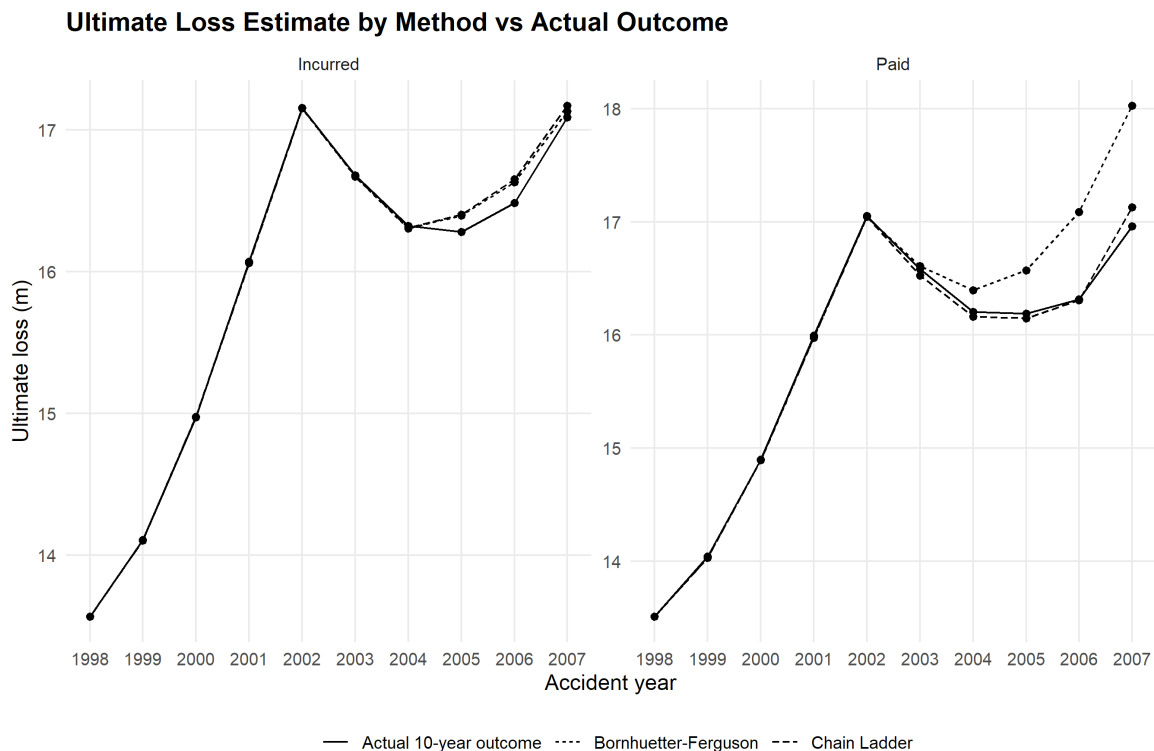
Figure 4 compares the Chain Ladder and Bornhuetter-Ferguson ultimate estimates against the retained 10-year outcome. The incurred estimates are close to the final incurred outcome on both

Table 3. Method comparison summary.

Basis	CL ultimate	BF ultimate	Actual	CL reserve	BF reserve	BF error vs actual
Paid	157.7m	160.1m	157.7m	18.9m	21.3m	2.4m
Incurred	159.1m	159.0m	158.7m	-0.7m	-0.7m	0.3m

CL: Chain Ladder. BF: Bornhuetter–Ferguson. Incurred reserve denotes development from the latest incurred position.

methods. On the paid basis, Bornhuetter-Ferguson is visibly higher for the most recent accident years, while Chain Ladder remains closer to the retained paid outcome in aggregate.

**Figure 4.** Ultimate loss estimates by method compared with retained 10-year outcomes.

4.3 Mack reserve uncertainty

Table 4. Mack Chain Ladder uncertainty summary.

Basis	Ultimate	Reserve	Mack SE	Lower 95%	Upper 95%	Actual
Paid	157.7m	18.9m	0.4m	18.2m	19.7m	157.7m
Incurred	159.1m	-0.7m	0.4m	-1.5m	0.1m	158.7m

The paid Mack result indicates a material remaining reserve, concentrated in the most recent accident years, with a relatively narrow interval around the aggregate estimate. The incurred result is different: the reserve estimate is slightly negative and the interval crosses zero. This supports the earlier interpretation that the incurred position was already close to fully developed at the valuation date.

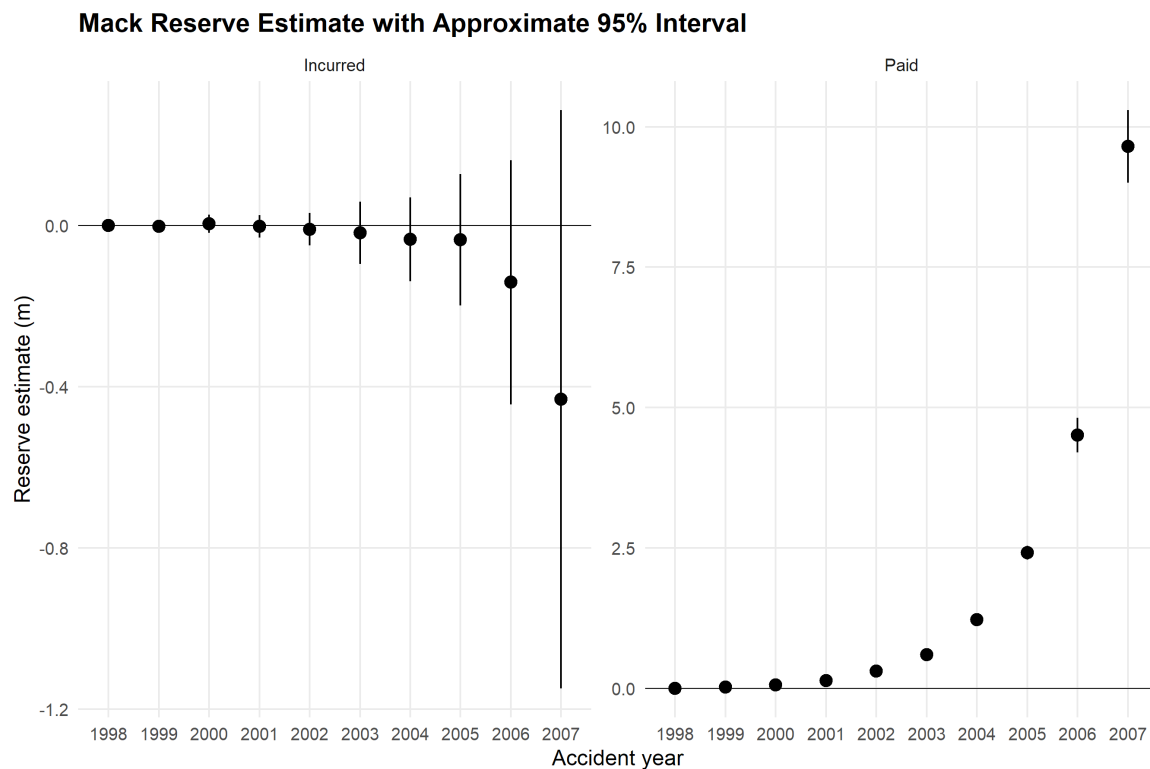


Figure 5. Mack reserve estimates with approximate 95% intervals by accident year.

The uncertainty pattern is consistent with the development stage of the accident years. Older years are almost fully developed and therefore have limited remaining reserve uncertainty. The largest paid reserve and uncertainty appear in the 2006 and 2007 accident years, where only the earliest development lags were available at the valuation date.

5 Limitations and further work

The analysis is based on aggregated Schedule P records rather than claim-level data. This is appropriate for triangle-based reserving, but it does not allow separate modelling of claim frequency, severity or settlement timing. The portfolio is also aggregated across companies and groups, so the results describe a broad private passenger auto portfolio rather than a single insurer's reserve position.

The retained full 10-year outcomes are used only as a retrospective validation benchmark. They are not used to fit the Chain Ladder or Mack models. For Bornhuetter-Ferguson, the expected loss ratio is selected from older observed experience to provide a stable prior; in a live reserving exercise, this assumption would usually be reviewed alongside business mix, pricing, claims inflation and underwriting changes.

The approximate Mack intervals should also be interpreted as model-based uncertainty around the Chain Ladder assumptions, not as complete reserve risk. They do not capture all sources of uncertainty, such as changes in claims handling, legal environment, inflation or exposure mix.

The close aggregate paid result is specific to this portfolio and valuation date. Offsetting errors across accident years can make total error small even when individual-year estimates are less accurate. A stronger evaluation would repeat the exercise at several historical valuation dates.

Further work could segment the data by company or business mix, compare other lines of business, test alternative expected loss ratio priors, and supplement Mack uncertainty estimates with bootstrap or overdispersed Poisson methods.

6 Conclusion

The project produced a coherent reserving workflow from Schedule P data through to reserve estimates and uncertainty analysis. The paid and incurred triangles give different but complementary views of the reserve position. Paid losses show material remaining development at the 2007 valuation date, while incurred losses are already close to ultimate and show modest reserve releases.

On the paid basis, Chain Ladder estimated an ultimate loss of 157.7m and a reserve of 18.9m, closely matching the retained 10-year paid outcome. Bornhuetter-Ferguson gave a more conservative paid reserve of 21.3m, driven by the expected loss ratio prior on immature accident years. Mack Chain Ladder estimated a paid reserve interval of 18.2m to 19.7m.

On the incurred basis, both Chain Ladder and Bornhuetter-Ferguson produced small negative incurred-development estimates, with Mack uncertainty crossing zero. This is consistent with the observed incurred development pattern and suggests that the incurred view at the valuation date was broadly aligned with the eventual 10-year incurred outcome.

The exercise links triangle construction, deterministic reserve estimation, prior-based comparisons and model-based uncertainty. The distinction between paid reserves and incurred development is central to interpreting the results.

References

- [1] Casualty Actuarial Society. *Loss Reserving Data Pulled from NAIC Schedule P*. Available at: <https://www.casact.org/publications-research/research/research-resources/loss-reserving-data-pulled-naic-schedule-p>
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